

Yuru Sun

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SUMMARY

PhD candidate in Monash University with strong expertise in statistical modelling, Bayesian methods and data analytics. Strong programming capability in R, python and MATLAB.

WORK EXPERIENCE

Teaching Associate

Monash Business School, Feb 2022 – July 2025

- Demonstrated statistical modelling and data analysis skills in dealing with longitudinal data in R;
- Helping with unit delivery with SQL and PowerBI;
- Delivered tutorials across undergraduate and postgraduate units, supporting students to interpret quantitative results and apply modelling tools to real datasets.
- Demonstrated reliable delivery under tight timelines and strong ability to communicate technical methods to non-technical audience;
- Recognised with Excellent Teaching Team Awards and consistently achieved top-tier student evaluation outcomes.
- Demonstrated strong planning and delivery skills through timely feedback during peak teaching periods while maintaining research progress.

Research Assistant (Modeller)

Monash Business School, Aug 2023 – Oct 2024

- Conducted empirical analysis on high-frequency bond market data to quantify market behaviour and assess the impact of regulatory and policy-related changes;
- Developed reproducible R workflows for data cleaning and diagnostics, extracting insights into data structure through visualisations and descriptive statistics to inform modelling choices when evaluating policy impacts.
- Built time-series econometrics model: linear regression, logistic regression with bootstrap inference, and translated the results into clear and decision-oriented summaries.

Research Assistant (Data Analyst)

Melbourne Business School, Aug 2022 – Jan 2023

- Automated large-scale high-frequency equity data retrieval and processing pipelines using Python and R for millisecond return of 500 stocks, improving efficiency and reproducibility of dataset construction;
- Cleaned and validated high-frequency time-series data, implemented quality checks, and produced analysis-ready datasets to support modelling and forecasting tasks;
- Collaborated with academic stakeholders to translate project requirements into structured datasets and deliverables.

Macroeconomics Research Analyst Internship

Liaowang Think Tank, July 2021 – Oct, 2021

- Analysed fiscal and monetary policy and evaluated their economic impacts on the public, producing evidence-based analysis to inform policy discussions.
- Produced written reports and presentation-ready materials for non-technical audiences, strengthening stakeholder communication and clarity of insights.

RESEARCH PROJECT EXPERIENCE

- Improvements on Market Risk Measures

Monash, Nov 2024 - present

 - Implemented non-linear optimisation skills to estimate model parameters within large-scale simulation and empirical frameworks.
 - Implemented numerical interpolation techniques, such as spline and polynomial interpolation, to construct continuous model inputs for simulation and optimisation.
 - Experience in processing and cleaning option datasets in R and MATLAB.
 - Designed Monte Carlo simulations under varying market conditions for scenario analysis.

- Modelling Financial Price Jumps

UniMelb & Monash, July 2022 - 2025

 - Designed advanced statistical modelling frameworks for complex time-series dynamics under uncertainty using Bayesian methods.
 - Built stochastic volatility models for financial returns and price jumps and designed scenario tests for model performance evaluation.
 - Mastered the parametric/nonparametric models for modelling economic states.
 - Processed, cleaned and analysed high-frequency data in R, Python and MATLAB.
 - Built predictive analytics pipelines to evaluate return distributions and quantify risk exposures under varying market regimes.
 - Implemented reliable version control workflows using Git and executed large-scale computational jobs on HPC clusters via SLURM.
 - Communicated complex quantitative results through clear visualizations, conference presentations, and technical reports for both academic and applied audiences.

- Optimal Probabilistic Forecast for Risk Management

Monash, March 2020 - 2022

 - Implemented advanced statistical modelling and estimation methods to construct probabilistic forecasts of financial returns.
 - Built reproducible workflows for numerical optimisation, model estimation, and probabilistic forecasting.
 - Applied probabilistic forecasts in a risk management context (e.g. VaR backtesting and Expected shortfall) and developed hedging strategies using real market data.

EDUCATION

- 2023 - 2026

PhD in Econometrics and Business Statistics - **Monash University**
- 2022 - 2023

PhD in Financial Econometrics - **University of Melbourne** (Commenced in 2022, transferred to Monash in 2023)
- 2021

Graduate Certificate in Economics Analysis at **University of Sydney**
- 2019 - 2020

Bachelor in Econometrics (1st class honors) at **Monash University** (GPA: 3.9/4.0)
- 2017 - 2019

Bachelor in Commerce (Finance and Econometrics) at **Monash University**

SKILLS

R, MATLAB, VBA, Python, Linux, Git, Excel, STATA, SQL, Power BI

HONOURS & AWARDS

- 2025 Excellent Teaching Team Awards for Business Forecasting, **Monash Business School**
- 2025 Prestigious International Conference Award, **Monash Business School**
- 2025 CFM/SoFiE Scholarship, **Capital Fund Management & Society for Financial Econometrics**
- 2024 Excellent Teaching Team Awards for Data Analysis in Business, **Monash Business School**
- 2024 Best Poster Runner-Up, **Time Series and Forecasting Symposium**, University of Sydney
- 2023 Co-funded Monash Graduate Scholarship, **Monash University**
- 2023 Monash Business School Top-Up Scholarship, **Monash University**
- 2022 Faculty of Business and Economics Graduate Research Scholarship, **University of Melbourne**
- 2021 Dean's Honour List (First Class), **Monash Business School**
- 2020 Econometrics Memorial Scholarship, **Monash University**
- 2020 Meritorious Winner, **Interdisciplinary Contest in Modelling (ICM)**